

Employee Dataset – Machine Learning Classification Tasks

Use the given Employee dataset and perform the following tasks.

- 1 Load the Employee dataset into a Pandas DataFrame and display the first 5 rows, last 5 rows, shape, column names, and basic information about the dataset.
- 2 Identify the target column for the classification problem. Clearly state what the target represents.
- 3 Check the dataset for missing values and duplicate rows. If any are present, handle them appropriately and explain your choice.
- 4 Identify the numerical and categorical columns in the dataset.
- 5 Examine all columns and identify whether any column is unnecessary for building the classification model. Remove only the columns that you can justify removing.
- 6 Encode the categorical columns so that they can be used by the machine learning algorithms.
- 7 Create a correlation matrix for the numerical/encoded features and display it using a heatmap. Analyze the relationships between the features and the target variable.
- 8 Create a visualization showing the distribution of the target variable (LeaveOrNot).
- 9 Create suitable visualizations to understand how numerical features such as Age, JoiningYear, PaymentTier, and ExperienceInCurrentDomain are related to LeaveOrNot.
- 10 Create suitable visualizations to analyze the relationship between the categorical features and LeaveOrNot.
- 11 Prepare the feature matrix X and target variable y. Make sure the target column is not included inside X.
- 12 Split the dataset into training and testing sets using a suitable train-test ratio and a fixed random_state.
- 13 Apply feature scaling to the features before training the KNN model. Explain why scaling is required for KNN.
- 14 Build a K-Nearest Neighbors (KNN) classification model and train it using the training data.
- 15 Try different values of K and observe how the model performance changes. Select a suitable K based on your observations.
- 16 Build a Gaussian Naive Bayes classification model using the same training and testing data.
- 17 Generate predictions using both KNN and Gaussian Naive Bayes models.
- 18 Evaluate both models using accuracy score, confusion matrix, and classification report.
- 19 Compare the KNN and Gaussian Naive Bayes results using the evaluation metrics and explain the differences you observe.